PROBLEM9:

PROBLEM STATEMENT:

Multi-Currency Travel Budget Analysis: Convert travel expenses using supplied exchange rates, incorporate conversion fees, compare planned spending against a budget, and identify the largest expense category

OBJECTIVE:

To develop a Python program that converts travel expenses from different currencies into a base currency using supplied exchange rates, includes conversion fees, compares total spending with the planned budget, and identifies the largest expense category.

PROGRAM:

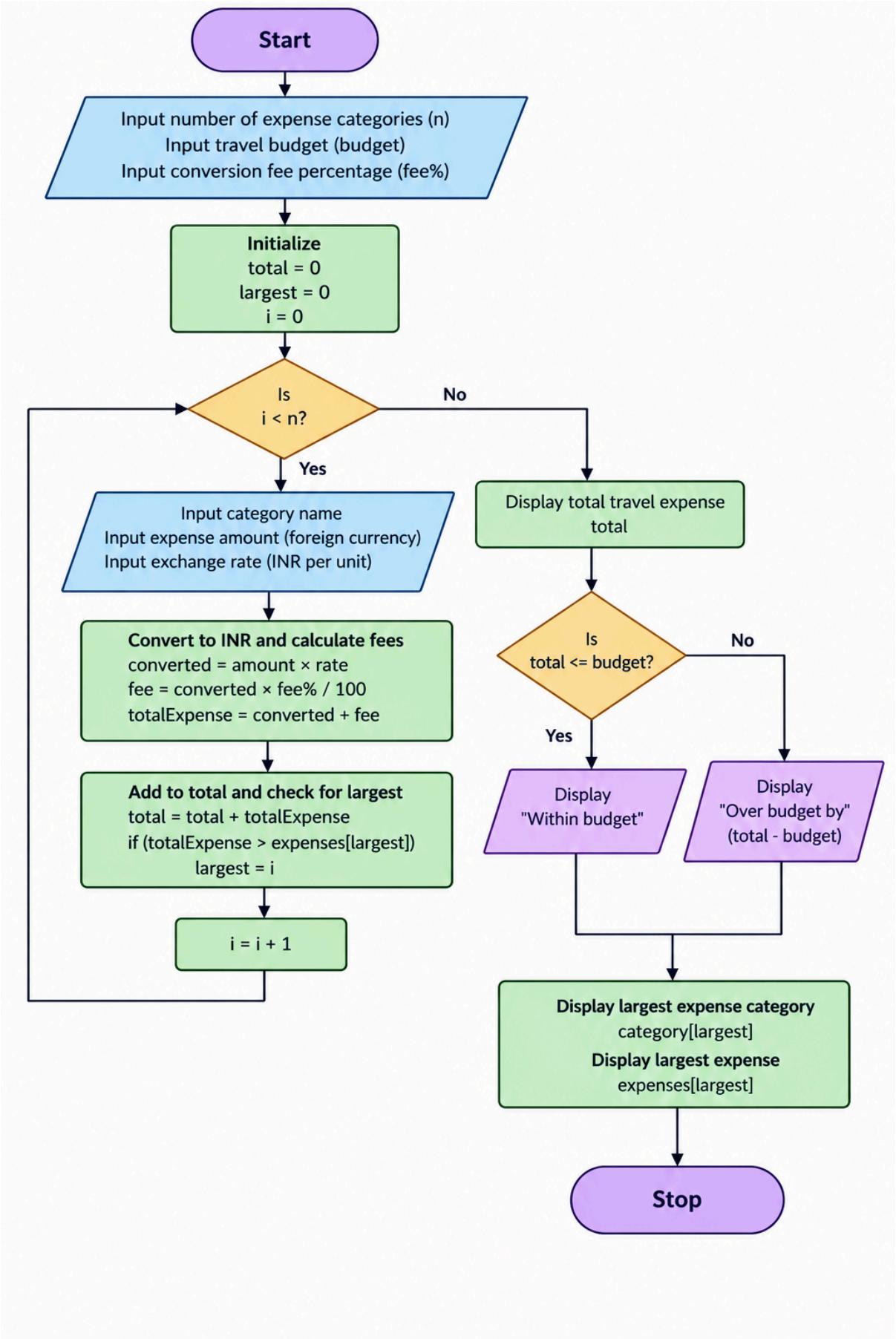
```
# Multi-Currency Travel Budget Analysisbudget = float(input("Enter travel budget in INR: "))n = int(input("Enter number of expenses: "))total_expense = 0category_totals = {}for i in range(n):    print("\nExpense", i + 1)    category = input("Enter expense category: ")    amount = float(input("Enter expense amount: "))    rate = float(input("Enter exchange rate (INR per unit): "))    fee_percent = float(input("Enter conversion fee percentage: "))converted_amount = amount * rate    fee = converted_amount * fee_percent / 100final_amount = converted_amount + fee    total_expense += final_amount    if category in category_totals:        category_totals[category] += final_amount    else:category_totals[category] = final_amount    print("Converted Amount: Rs.", round(converted_amount, 2))    print("Conversion Fee: Rs.", round(fee, 2))    print("Total Expense: Rs.", round(final_amount, 2))print("\n----- TRAVEL BUDGET SUMMARY -----")print("Planned Budget: Rs.", round(budget, 2))print("Total Spending: Rs.", round(total_expense, 2))
```

ALGORITHM:

1. Start.
2. Input the planned travel budget.
3. Input the number of travel expenses.
4. For each expense, enter its category, amount, exchange rate, and conversion fee percentage.
5. Convert the expense into INR using the exchange rate.
6. Calculate the conversion fee and add it to the converted amount.
7. Add the final expense to the total spending and the corresponding category total.
8. Compare total spending with the planned budget.
9. Identify the category with the highest total expense.

10. Display the budget status, remaining budget or excess amount, and largest expense category.
11. Stop.

FLOWCHART:


Start
Input number of expense categories (n)
Input travel budget (budget)
Input conversion fee percentage (fee%)
Initialize
total = 0
largest = 0
i = 0
Is
i < n?
No
Yes
Input category name
Input expense amount (foreign currency)
Input exchange rate (INR per unit)
Convert to INR and calculate fees
converted = amount × rate
fee = converted × fee% / 100
totalExpense = converted + fee
Add to total and check for largest
total = total + totalExpense
if (totalExpense > expenses[largest])
largest = i
i = i + 1
Display total travel expense
total
Is
total <= budget?
No
Yes
Display
"Within budget"
Display
"Over budget by"
(total - budget)
Display largest expense category
category[largest]
Display largest expense
expenses[largest]
Stop

PSEUDOCODE:

```
START
INPUT budget
INPUT number_of_expenses
total_expense = 0

FOR each expense:
    INPUT category, amount, rate, fee_percent
    converted = amount * rate
    fee = converted * fee_percent / 100
    final_amount = converted + fee
    total_expense = total_expense + final_amount
    ADD final_amount to category total
    DISPLAY converted, fee, final_amount
END FOR

IF total_expense <= budget:
    DISPLAY "Within Budget"
    DISPLAY budget - total_expense
ELSE:
    DISPLAY "Budget Exceeded"
    DISPLAY total_expense - budget
END IF

FIND category with maximum total
DISPLAY largest category and its amount
STOP
```

OUTPUT:

Example input: Budget ₹100,000; two expenses

```
----- TRAVEL BUDGET SUMMARY -----
Planned Budget: Rs. 100000.0
Total Spending: Rs. 86700.0
Status: Within Budget
Remaining Budget: Rs. 13300.0
Largest Expense Category: Accommodation
Amount: Rs. 61200.0
```

The sample assumes accommodation costs ₹61,200 and another category costs ₹25,500. Actual output depends on the user's inputs.